SHRUTI SANJAY HALDE

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PROFESSIONAL SUMMARY

Data-driven professional with expertise in SQL, Power BI, Python, and AI/ML. Skilled in building ML pipelines, large language models, and data dashboards. Proven in transforming datasets, optimizing pipelines, and automating reporting workflows to drive business insights. Recently interned at Trent Ltd., enhancing model performance and automating processes. Adept at solving complex problems through data analysis and machine learning techniques. Committed to leveraging data to drive actionable insights and optimize business strategies.

EDUCATION

University of Southern California **Aug 2024 - Jun 2026**

Master of Science, Computer Science

- **GPA:** 3.52/4
- **Coursework:** Foundations of Artificial Intelligence, Applied Natural Language Processing, Deep Learning and its Applications, Analysis of Algorithms, Database Systems

MCT's Rajiv Gandhi Institute of Technology **Aug 2019 - Jun 2023**

Bachelor, Computer Engineering

- **GPA:** 9.76/10
- **Achievements:** Ranked 1st in University
- **Coursework:** Machine Learning, Database Management Systems, Natural Language Processing, Artificial Intelligence

INTERNSHIPS

Trent Ltd. - A TATA Enterprise **May 2025 - Jun 2025**

Data Analyst Intern *India*

- Developed an Inventory Management dashboard using Power BI, enhancing real-time stock tracking, inventory turnover insights, and informed operational decision-making.
- Automated data pipelines and reporting workflows by integrating Power BI Report Builder, Power Automate, SharePoint, and Microsoft Forms, which streamlined report generation and reduced manual errors.
- Executed comprehensive SQL queries in SSMS to extract, clean, and transform large datasets, and built Python scripts for anomaly detection and ad hoc analysis to support data-driven decisions.

Suvidha Foundation **Jul 2023 - Aug 2023**

Machine Learning Intern *India*

- Constructed a text summarization model using Transformers and Seq2Seq in Python, enhancing the NLP pipeline for improved context retention and achieving a 60% reduction in summary length.
- Optimized pre-trained NLP models on custom datasets through fine-tuning to improve task-specific performance and address domain-specific requirements.
- Conducted rigorous error analysis and model evaluation using ROUGE and BLEU metrics, leading to a 12% improvement in the model's F1-score through iterative refinements.

TCR Innovation **Jun 2021 - Aug 2021**

Data Science with ML & Python Intern *India*

- Collaborated on real-world datasets to implement machine learning models using Pandas, NumPy, and TensorFlow, identifying trends that informed business insights.
- Executed data preprocessing and exploratory data analysis (EDA), improving dataset usability and model performance by optimizing feature selection.
- Delivered findings through compelling visualizations and presentations, demonstrating strong skills in data storytelling and stakeholder communication.

PROJECTS

Multiple Disease Prediction using Machine Learning **Jan 2023 - May 2023**

- Built classification models (Logistic Regression, SVM, Random Forest) on medical data; performed EDA, feature selection, and evaluation using F1-score and confusion matrix.
- Published results in IC-ICN 2023, demonstrating the diagnostic potential of predictive healthcare modeling.

QueryBot - Natural Language to SQL Generation **Jan 2025 - May 2025**

USC Academic Project

- Developed a semantic parser using T5, SQL-BART, and SmBoP to translate user queries to SQL; fine-tuned models on the Spider dataset.
- Achieved 12% accuracy improvement through schema linking; deployed via Streamlit with logical and execution accuracy tracking.

Legal Question Answering System **Jan 2025 - May 2025**
USC Academic Project
- Fine-tuned LLaMA and Mistral LLMs for legal QA tasks using RAG and Gemini-generated prompts on legal corpora.
- Integrated QLORA and Flash Attention with vector databases for efficient citation-aware response generation.

Road Accident Dashboard **Jun 2025 - Jul 2025**
Personal Project
- Created an interactive dashboard in Power BI and Tableau to analyze 2021-2022 road accident data across KPIs, vehicle/road types, and time of day.
- Used SQL and Excel for data prep and DAX for monthly trend comparisons and geospatial filtering.

Attendance System Using Facial Recognition **Aug 2021 - May 2022**
- Built a real-time attendance system using Python and OpenCV; automated facial recognition and attendance logging.
- Connected to MySQL backend for secure data storage and deployed for live pilot use in classrooms.

Weather Forecasting Android App **Aug 2024 - Dec 2024**
USC Academic Project
- Built a weather app integrating Highcharts, IPinfo, and Google Places API to deliver location-based forecasts.
- Hosted backend on AWS and implemented real-time interactive visualizations.

Al Flappy Bird (NEAT Algorithm) **Jun 2024 - Jul 2024**
Individual Project
- Designed a neuroevolution-based RL agent using the NEAT algorithm to learn gameplay strategies in Flappy Bird.
- Implemented and trained evolving neural networks to maximize decision accuracy.

SKILLS

- **Programming & Query Languages**: Python, SQL, R (learning), C, Java (basic)
- **Data Science & ML**: Data Analysis, EDA, Feature Engineering, Classification, Regression, Clustering, Model Evaluation, Statistical Modeling, Statistics, Predictive Analytics, Descriptive Statistics
- **Data Visualization & Reporting**: Power BI (DAX), Tableau, Excel (Pivot Tables, VLOOKUP), Google Sheets, Matplotlib, Seaborn, Data Visualization, Power Query
- **Libraries & Frameworks**: Pandas, NumPy, Scikit-learn, TensorFlow, Hugging Face Transformers, LangChain
- **Tools & Platforms**: SQL Server (SSMS), MySQL, GCP, AWS (learning), Azure, Power Automate
- **Key Concepts**: Data Cleaning, ETL Pipelines, KPI Dashboards, Data Modeling, NLP, LLM Fine-tuning, Secure Data Practices
- **Soft Skills**: Communication, Data Storytelling, Problem Solving, Collaboration, Time Management

CERTIFICATIONS

- Google Data Analytics
- Udemy: Python Bootcamp
- Elements of Al - University of Helsinki
- Google Project Management
- NPTEL – AI: Search Methods for Problem Solving
- DeepLearning. Al - Mathematics for ML and Data Science

EXTRA CURRICULAR ACTIVITIES

Athena Hacks **2025**
Mentor
- Guided participants in Al project ideation and troubleshooting

AME Lab, USC
Research Assistant
- Brief role supporting Al-aerospace integration

CESS, Undergraduate College
Event Manager
- Organized tech fests, managed sponsor outreach

Society of Women Engineers (SWE) **2024 - 2025**
Member